SevaOne: High-Level Architecture

Current implementation. Drawn from the repository and the running stack.

Service · Request · Event or callback · Disabled or off-box

Data store · WEB · MOBILE · AI · Which channel uses an outside system

1. CLIENTS AND CHANNELS

Citizen web portal
React 19 · Vite · TanStack Router
Tailwind 4 · Zustand · i18n

Operator console
Assisted / kiosk mode
Same React app

Admin panel
Government admin console
React 19 SPA · analytics, catalogue

Mobile app
Flutter · Android
in.vismai.sp.sevaone

HTTPS · HTTPS · HTTPS

2. EDGE AND SECURITY

NGINX reverse proxy
TLS 1.2 / 1.3 · security headers (CSP, frame deny) · routes /api to the gateway, everything else to the SPA

/api/v1/*

3. GATEWAY AND IDENTITY

HTTPS · JWT

WSO2 API Manager 4.7
Gateway :8243 · API products sevaone-citizen and sevaone
API keys · gateway secret stamped on every call

WSO2 Identity Server 7.3
SCIM2 · OAuth2 JWT-bearer (RFC 7523)
Issuers: citizen · operator · government admin

forwarded + secret · SCIM2 + JWT

department callbacks

4. APPLICATION SERVICES

.NET 10 backend for frontend (SevaOne.Services)
Dapper + Npgsql (no ORM) · FluentValidation · Serilog · OpenTelemetry · health checks

Services
Controllers + middleware
Auth · Catalogue · Applications · Operator · Department · Gov admin · AI chat

Business
Services + validators
Apply, track and department-handoff logic

Infrastructure
Clients + repositories
Postgres functions · WSO2 · DigiLocker · SMS · AI clients

Domain
Models + error codes
Stable SEVA-* error contract

Background: OutboxDispatcher reads the outbox table and publishes events to Kafka · Flyway runs the database migrations at start-up

ai-chat

AI services (GPU server)
sevaone-rag: agentic RAG, :8000
vLLM model server · Bhashini Kannada
ai-admin :8090 · ai-monitor :8010
(reached over SSH tunnels)

AI

Mobile backend (FastAPI)
Mobile sign-in (OTP) · catalogue
Applications · documents · DigiLocker
Push notifications · grievances
Python 3 · psycopg

MOBILE

Npgsql · sessions · outbox · retrieve · psycopg

5. DATA AND STREAMING

PostgreSQL 18
6 schemas: discovery, portal, orchestration, status, audit, notification. Function-only access.

Redis 8
Sessions and token cache
Data-protection keys

Apache Kafka 4.3
Transactional outbox events
Published by OutboxDispatcher

MinIO
Object storage
Disabled for now

Vector index
FAISS or Qdrant
Services and knowledge passages

PostgreSQL and files
Mobile catalogue, sessions
documents · local or object storage

outbound calls and redirects

6. EXTERNAL ECOSYSTEM

Department systems
Nadakacheri (Revenue)
Department Hub · AJSK

WEB

DigiLocker
CeG gateway
Aadhaar OTP, eAadhaar

WEB · MOBILE

SMS gateway
One-time passwords
Mobile Seva

WEB · MOBILE

Public catalogue sites
Sakala · ESCOM
Seva Sindhu catalogue

WEB · MOBILE

ServicePlus
Live service catalogue and applications

MOBILE

Firebase
Push messages (FCM)

MOBILE

Bhashini and LLM
Kannada speech, translation
vLLM language model

AI · MOBILE